

The j -formula for $\lambda = (2, 2, 2, 1^{n-6})$ via branching reduction to $(2, 2, 1^{n-6})$

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Abstract

We give a fully rigorous proof of the j -formula at $j = n - 2$ for $\lambda = (2, 2, 2, 1^{n-6})$ at every $n \geq 9$, completing the May-14 sketch. The strategy is a *branching reduction*: the $H_q(S_{n-2})$ -equivariant isomorphism $\Phi : V_\mu|_{n-2} \xrightarrow{\sim} E_{n-1}^+|_{V_\lambda|_n}$ (where $\mu = (2, 2, 1^{n-6})$) reduces three of the four staircase factors of $B_{n-2}^{(\lambda)}$ to the May-13 j -formula at μ , leaving only two extra factors $(T_{n-3}+1)(T_{n-2}+1)$ to analyze explicitly. We obtain

$$B_{n-2}^{(\lambda)} V_\lambda = \mathbb{Q}(q) \cdot (T_{n-3}+1)(T_{n-2}+1) \Phi(u^{(\mu)}),$$

where $u^{(\mu)} \in V_\mu$ is the May-13 generator of $B_{n-3}^{(\mu)} V_\mu$. A direct case analysis on the eight supports of $\Phi(u^{(\mu)})$ shows the right-hand side is 1-dimensional, contained in E_1^- , and supported on the eight SYTs of $S_3^{(n)}$ from the recursive set defined in `2026-05-14-2a-1n-family.tex`.

As a corollary, $\Pi^{S_n}|_{V_{(2,2,2,1^{n-6})}} = 0$ unconditionally for every $n \geq 9$ — the second non-hook family in the rank-zero program, now fully structurally proved.

1 Setup

We use the conventions of `2026-05-13-non-hook-22-1n.tex` and `2026-05-14-2a-1n-family.tex` (which we abbreviate **May-13** and **May-14**). $H_q(S_n)$ is the Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ satisfying $T_i^2 = (q-1)T_i + q$ and the braid relations. V_λ is the seminormal cell module of $H_q(S_n)$ for $\lambda \vdash n$, with basis $\{v_T\}$ indexed by standard Young tableaux (SYTs). We write

$$R'_k = (T_{k-1}+1)(T_{k-2}+1) \cdots (T_1+1), \quad B_j = R'_j R'_{j+1} \cdots R'_n, \quad \Pi^{S_n} = B_2.$$

$E_i^\pm$ is the $\pm$ -eigenspace of T_i (eigenvalues q for $+$, -1 for $-$).

Throughout, $n \geq 9$ and

$$\lambda := (2, 2, 2, 1^{n-6}) \vdash n, \quad \mu := (2, 2, 1^{n-6}) \vdash n-2.$$

$\ell(\lambda) = n-3$, so the j -formula is at $j = \ell + 1 = n-2$ and $B_{n-2}^{(\lambda)} = R'_{n-2} R'_{n-1} R'_n$.

1.1 Seminormal basis of V_λ

By May-14 Lemma 1.2, SYTs of λ are in bijection with triples (a_1, a_2, a_3) with $2 \leq a_1 < a_2 < a_3 \leq n$ satisfying the validity conditions $a_2 > b_1$, $a_3 > b_2$, where b_i is the i -th smallest element of $\{2, \dots, n\} \setminus \{a_1, a_2, a_3\}$. We write $v_{(a_1, a_2, a_3)} := v_T$ for the corresponding SYT.

By May-14 Lemma 1.5, T_1 acts diagonally on V_λ :

$$E_1^+ = \text{span}\{v_{(2, a_2, a_3)} : \text{valid}\}, \quad E_1^- = \text{span}\{v_{(a_1, a_2, a_3)} : a_1 \geq 3, \text{ valid}\}.$$

1.2 Seminormal basis of V_μ

By May-13 Lemma 2.1, SYTs of $\mu = (2, 2, 1^{m-4})$ at m letters are in bijection with pairs (α, γ) satisfying $2 \leq \alpha < \gamma \leq m$ and $(\alpha, \gamma) \neq (2, 3)$. We use the convention $v_{(\alpha, \gamma)}^{(\mu)} := v_{T_{(\alpha, \gamma)}}^{(\mu)}$ in $V_\mu|_m$. Here $m = n - 2$, so (α, γ) ranges in $\{2, \dots, n - 2\}^2$.

1.3 Phase A and the May-13 j -formula

We rely on two prior results:

Theorem 1 (Phase A for λ , May-14 Proposition 3.1). $R'_n V_\lambda = E_{n-1}^+$, of dimension $(n-2)(n-5)/2$.

Theorem 2 (May-13 main theorem; 2026-05-13-non-hook-22-1n.tex Theorem 2.5). For $m \geq 6$, $\mu = (2, 2, 1^{m-4}) \vdash m$ at length $\ell(\mu) = m - 2$ and $j(\mu) = m - 1$,

$$B_{m-1}^{(\mu)} V_\mu = R_{m-1}' R_m^{(\mu)} V_\mu \subseteq E_1^-|_{V_\mu},$$

of dimension 1, with support exactly on the 4 SYTs $\{(m-3, m-2), (m-3, m-1), (m-2, m), (m-1, m)\}$. Moreover, by May-13 Phase A endpoint (Proposition 3.1 of that paper), $R_m^{(\mu)} V_\mu = E_{m-1}^+|_{V_\mu}$, of dimension $m - 3$.

We apply Theorem 2 at $m = n - 2$. Setting $\hat{u}^{(\mu)} \in V_\mu$ to be a fixed generator of $B_{n-3}^{(\mu)} V_\mu$ (unique up to nonzero scalar), May-13 gives the explicit form

$$\hat{u}^{(\mu)} = A v_{(n-5, n-4)}^{(\mu)} + B v_{(n-5, n-3)}^{(\mu)} + C v_{(n-4, n-2)}^{(\mu)} + D v_{(n-3, n-2)}^{(\mu)}, \quad A, B, C, D \in \mathbb{Q}(q)^\times. \quad (1)$$

2 The main theorem

Theorem 3 (j -formula at $\lambda = (2, 2, 2, 1^{n-6})$, $n \geq 9$). $B_{n-2}^{(\lambda)} V_\lambda = R_{n-2}' R_{n-1}' R_n' V_\lambda$ is 1-dimensional, contained in $E_1^-|_{V_\lambda}$, with support exactly the 8 SYTs of $S_3^{(n)}$:

$$S_3^{(n)} = \left\{ (n-5, n-4, n-3), (n-5, n-4, n-2), (n-5, n-3, n-1), (n-5, n-2, n-1), \right. \\ \left. (n-4, n-3, n), (n-4, n-2, n), (n-3, n-1, n), (n-2, n-1, n) \right\}.$$

Corollary 4 (Rank-zero, structurally). $\Pi^{S_n}|_{V_{(2,2,2,1^{n-6})}} = 0$ for every $n \geq 9$.

Proof. Theorem 3 gives $B_{n-2}^{(\lambda)} V_\lambda \subseteq E_1^-$. Factor $\Pi^{S_n} = R_2' \cdots R_{n-3}' \cdot B_{n-2}^{(\lambda)}$. The leftmost factor R_{n-3}' ends in (T_1+1) , which annihilates E_1^- . $\square$

The remainder of the paper proves Theorem 3.

3 The branching isomorphism Φ

The subalgebra of $H_q(S_n)$ generated by $\{T_1, \dots, T_{n-3}\}$ is canonically isomorphic to $H_q(S_{n-2})$ (permuting letters $\{1, \dots, n-2\}$). Since T_{n-1} commutes with each T_i for $i \leq n-3$ (index gap ≥ 2), the eigenspace E_{n-1}^+ is preserved by the $H_q(S_{n-2})$ -action and is therefore an $H_q(S_{n-2})$ -module.

Definition 5 (The map Φ). For each SYT T' of μ on letters $\{1, \dots, n-2\}$ with column-2 entries (α, γ) , define two SYTs T_A, T_B of λ on letters $\{1, \dots, n\}$:

- T_A : extend T' by placing $n-1$ at $(3, 2)$ and n at the new bottom row $(n-3, 1)$.
- T_B : extend T' by placing n at $(3, 2)$ and $n-1$ at $(n-3, 1)$.

Both are valid SYTs of λ (since $n-1, n$ exceed all other entries, the column- and row-increase conditions are trivially satisfied). Letting $d = c(n) - c(n-1)$ be the axial distance in T_A (where $c(r, c) = c - r$ denotes content), we have $d = (-1) - (4 - n) = n - 5 \geq 4$ for $n \geq 9$.

Define

$$\Phi(v_{T'}^{(\mu)}) := \text{the } T_{n-1} + q\text{-eigenvector in the 2-block } \{v_{T_A}, v_{T_B}\} \subset V_\lambda,$$

normalized by setting the coefficient of v_{T_A} to 1 (i.e., $\Phi(v_{T'}^{(\mu)}) = v_{T_A} + \tau(d)v_{T_B}$ with $\tau(d)$ the Hoefsmit ratio at axial distance d , $\tau(d) \in \mathbb{Q}(q)^\times$ for $|d| \geq 2$). Extend Φ linearly to all of $V_\mu|_{n-2}$.

We write $w_\Phi(\alpha, \gamma) := \Phi(v_{(\alpha, \gamma)}^{(\mu)})$.

Lemma 6 (Φ is $H_q(S_{n-2})$ -equivariant). *For every $1 \leq i \leq n-3$ and every $v \in V_\mu$, $T_i \Phi(v) = \Phi(T_i v)$.*

Proof. By linearity, suffices to check $v = v_{T'}^{(\mu)}$. The positions of $i, i+1$ in T_A and T_B are identical to the positions in T' (since $i, i+1 \leq n-2$ and the cells of T_A, T_B holding entries $\leq n-2$ agree with T'). Three cases for the T_i -action on T' :

(a) **Same row.** $T_i v_{T'}^{(\mu)} = q v_{T'}^{(\mu)}$. Then $T_i v_{T_A} = q v_{T_A}$ and $T_i v_{T_B} = q v_{T_B}$ (same positions of $i, i+1$). Hence $T_i \Phi(v_{T'}^{(\mu)}) = T_i(v_{T_A} + \tau v_{T_B}) = q(v_{T_A} + \tau v_{T_B}) = q\Phi(v_{T'}^{(\mu)}) = \Phi(T_i v_{T'}^{(\mu)})$.

(b) **Same column.** Symmetric to (a), eigenvalue -1 .

(c) **T_i 2-block on V_μ at axial distance d_i .** The block partner is $s_i T'$. The $H_q(S_n)$ -action on V_λ has T_i -2-blocks at the SYTs $\{T_A, s_i T_A\}$ and $\{T_B, s_i T_B\}$, each at the *same* axial distance d_i (the cells of $i, i+1$ are at identical positions across the four SYTs). Since T_i and T_{n-1} commute, the four-dimensional subspace spanned by $\{v_{T_A}, v_{T_B}, v_{s_i T_A}, v_{s_i T_B}\}$ is preserved by both, and the actions decouple:

- T_{n-1} : 2-blocks $\{v_{T_A}, v_{T_B}\}$ and $\{v_{s_i T_A}, v_{s_i T_B}\}$, axial distance $n-5$ in each.
- T_i : 2-blocks $\{v_{T_A}, v_{s_i T_A}\}$ and $\{v_{T_B}, v_{s_i T_B}\}$, axial distance d_i in each.

By Hoefsmit, both T_{n-1} -2-blocks have the same coefficient matrix (same axial distance), so $\Phi(v_{T'}^{(\mu)}) = v_{T_A} + \tau v_{T_B}$ and $\Phi(v_{s_i T'}^{(\mu)}) = v_{s_i T_A} + \tau v_{s_i T_B}$ with the same τ . Similarly both T_i -2-blocks have the same coefficient matrix. Write $T_i v_{T_A} = a v_{T_A} + b v_{s_i T_A}$ and $T_i v_{T_B} = a v_{T_B} + b v_{s_i T_B}$ (same a, b). Then

$$\begin{aligned} T_i \Phi(v_{T'}^{(\mu)}) &= T_i(v_{T_A} + \tau v_{T_B}) = a v_{T_A} + b v_{s_i T_A} + \tau(a v_{T_B} + b v_{s_i T_B}) \\ &= a(v_{T_A} + \tau v_{T_B}) + b(v_{s_i T_A} + \tau v_{s_i T_B}) \\ &= a \Phi(v_{T'}^{(\mu)}) + b \Phi(v_{s_i T'}^{(\mu)}) = \Phi(a v_{T'}^{(\mu)} + b v_{s_i T'}^{(\mu)}) = \Phi(T_i v_{T'}^{(\mu)}). \end{aligned} \quad \square$$

Lemma 7 (Φ is an iso onto E_{n-1}^+). $\Phi : V_\mu|_{n-2} \xrightarrow{\sim} E_{n-1}^+|_{V_\lambda|_n}$ is a $\mathbb{Q}(q)$ -linear isomorphism.

Proof. Injectivity. Distinct T', T'' of μ give disjoint pairs $\{T_A, T_B\}$ and $\{T'_A, T'_B\}$ (the μ -shape is recovered by deleting the cells of $n-1, n$, which is unambiguous). Hence $\Phi(v_{T'}^{(\mu)})$ and $\Phi(v_{T''}^{(\mu)})$ have disjoint seminormal supports, and the family $\{\Phi(v_{T'}^{(\mu)})\}_{T'}$ is linearly independent.

Image contained in E_{n-1}^+ . Each $\Phi(v_{T'}^{(\mu)})$ is by construction a $T_{n-1} + q$ -eigenvector.

Dimension count. $\dim V_\mu = \frac{m(m-3)}{2} = \frac{(n-2)(n-5)}{2}$ (May-13 Lemma 2.4), equal to $\dim E_{n-1}^+|_{V_\lambda}$ by Theorem 1. $\square$

Lemma 8 (Φ preserves E_1^-). $\Phi(E_1^-|_{V_\mu}) \subseteq E_1^-|_{V_\lambda}$, and $\Phi(E_1^+|_{V_\mu}) \subseteq E_1^+|_{V_\lambda}$.

Proof. T_1 is in the $H_q(S_{n-2})$ -subalgebra and Φ is T_1 -equivariant (Lemma 6). Hence $T_1 v = \pm v$ implies $T_1 \Phi(v) = \pm \Phi(v)$. $\square$

4 The reduction formula

Theorem 9 (Reduction). *Set $S := (T_{n-4}+1)(T_{n-5}+1) \cdots (T_1+1) \in H_q(S_{n-3})$, and recall $R'_{n-2} = (T_{n-3}+1)S$. Then*

$$B_{n-2}^{(\lambda)} V_\lambda = (T_{n-3}+1)(T_{n-2}+1) \Phi(B_{n-3}^{(\mu)} V_\mu). \quad (2)$$

In particular, $\dim B_{n-2}^{(\lambda)} V_\lambda \leq 1$.

Proof. We compute step by step.

Step 1. $R'_n V_\lambda = E_{n-1}^+ = \Phi(V_\mu)$ by Theorem 1 and Lemma 7.

Step 2. $R'_{n-2} \in H_q(S_{n-2})$ uses only $T_1, \dots, T_{n-3}$. By Lemma 6, Φ intertwines this action:

$$R'_{n-2} \Phi(V_\mu) = \Phi(R'^{(\mu)}_{n-2} V_\mu) = \Phi(E_{n-3}^+|_{V_\mu}),$$

the last equality by May-13 Phase A (Theorem 2).

Step 3. $R'_{n-1} = (T_{n-2}+1)R'_{n-2}$, so

$$R'_{n-1} R'_n V_\lambda = (T_{n-2}+1) R'_{n-2} E_{n-1}^+ = (T_{n-2}+1) \Phi(E_{n-3}^+|_{V_\mu}).$$

Step 4. $R'_{n-2} = (T_{n-3}+1)S$ as in the theorem statement. Since S uses generators $T_1, \dots, T_{n-4}$ and T_{n-2} has index gap ≥ 2 from each, S commutes with T_{n-2} , hence with $(T_{n-2}+1)$. Therefore

$$R'_{n-2} (T_{n-2}+1) = (T_{n-3}+1) S (T_{n-2}+1) = (T_{n-3}+1) (T_{n-2}+1) S.$$

Step 5. $S \in H_q(S_{n-3}) \subset H_q(S_{n-2})$ is Φ -equivariant (Lemma 6), and $S = R'^{(\mu)}_{n-3}$ as an element of $H_q(S_{n-2})$. Hence

$$S \Phi(E_{n-3}^+|_{V_\mu}) = \Phi(R'^{(\mu)}_{n-3} E_{n-3}^+|_{V_\mu}) = \Phi(R'^{(\mu)}_{n-3} R'^{(\mu)}_{n-2} V_\mu) = \Phi(B_{n-3}^{(\mu)} V_\mu).$$

Combining. Therefore

$$\begin{aligned} B_{n-2}^{(\lambda)} V_\lambda &= R'_{n-2} R'_{n-1} R'_n V_\lambda = R'_{n-2} (T_{n-2}+1) \Phi(E_{n-3}^+|_{V_\mu}) \\ &= (T_{n-3}+1)(T_{n-2}+1) S \Phi(E_{n-3}^+|_{V_\mu}) \\ &= (T_{n-3}+1)(T_{n-2}+1) \Phi(B_{n-3}^{(\mu)} V_\mu). \end{aligned}$$

By Theorem 2, $\dim B_{n-3}^{(\mu)} V_\mu = 1$, so $\dim \Phi(B_{n-3}^{(\mu)} V_\mu) = 1$ (as Φ is injective) and the right-hand side has dimension at most 1. $\square$

Corollary 10 (E_1^- containment). $B_{n-2}^{(\lambda)} V_\lambda \subseteq E_1^-|_{V_\lambda}$.

Proof. By May-13, $B_{n-3}^{(\mu)} V_\mu \subseteq E_1^-|_{V_\mu}$. By Lemma 8, $\Phi(B_{n-3}^{(\mu)} V_\mu) \subseteq E_1^-|_{V_\lambda}$. The generators T_{n-3} and T_{n-2} commute with T_1 (index gap ≥ 2 for $n \geq 6$), so $(T_{n-3}+1)(T_{n-2}+1)$ preserves each T_1 -eigenspace. In particular, E_1^- is preserved. Apply Theorem 9. $\square$

5 Computing the lifted image

To complete the proof of Theorem 3, we must show:

1. $(T_{n-3}+1)(T_{n-2}+1)\Phi(\hat{u}^{(\mu)}) \neq 0$ (so the image is exactly 1-dimensional, not 0).
2. Its seminormal support equals $S_3^{(n)}$ from Theorem 3, with all 8 coefficients nonzero.

Recall (1): $\hat{u}^{(\mu)} = Av_{(n-5,n-4)}^{(\mu)} + Bv_{(n-5,n-3)}^{(\mu)} + Cv_{(n-4,n-2)}^{(\mu)} + Dv_{(n-3,n-2)}^{(\mu)}$, with $A, B, C, D \neq 0$. Applying Φ :

$$\Phi(\hat{u}^{(\mu)}) = Aw_{\Phi}(n-5, n-4) + Bw_{\Phi}(n-5, n-3) + Cw_{\Phi}(n-4, n-2) + Dw_{\Phi}(n-3, n-2), \quad (3)$$

where each $w_{\Phi}(\alpha, \gamma) = v_{(\alpha, \gamma, n-1)} + \tau v_{(\alpha, \gamma, n)}$ ($\tau \in \mathbb{Q}(q)^{\times}$ at axial distance $n-5 \geq 4$).

5.1 Step (a): apply $(T_{n-2}+1)$

We compute T_{n-2} on each of the 8 seminormal supports of $\Phi(\hat{u}^{(\mu)})$, namely $v_{(\alpha, \gamma, \delta)}$ for $(\alpha, \gamma) \in S_2^{(n-2)}$ and $\delta \in \{n-1, n\}$. Note T_{n-2} involves the entries $n-2$ and $n-1$.

Lemma 11 (Case analysis of T_{n-2}). *For $(\alpha, \gamma) \in S_2^{(n-2)}$ and $\delta \in \{n-1, n\}$:*

- (a) $(\alpha, \gamma, \delta) = (n-5, n-4, n)$ or $(n-5, n-3, n)$: $n-2$ and $n-1$ are both in column 1 at adjacent rows. $T_{n-2}v = -v$.
- (b) $(\alpha, \gamma, \delta) = (n-4, n-2, n-1)$ or $(n-3, n-2, n-1)$: $n-2, n-1$ are at $(2, 2), (3, 2)$ (same column 2). $T_{n-2}v = -v$.
- (c) $(\alpha, \gamma, \delta) = (n-5, n-4, n-1)$ or $(n-5, n-3, n-1)$: $n-1$ at $(3, 2)$, $n-2$ at $(n-4, 1)$. T_{n-2} acts as a 2-block at axial distance $d = n-6$ coupling $v_{(\alpha, \gamma, n-1)} \leftrightarrow v_{(\alpha, \gamma, n-2)}$.
- (d) $(\alpha, \gamma, \delta) = (n-4, n-2, n)$ or $(n-3, n-2, n)$: $n-2$ at $(2, 2)$ (content 0), $n-1$ at $(n-3, 1)$ (content $4-n$). T_{n-2} acts as a 2-block at axial distance $d = 4-n$ coupling $v_{(\alpha, n-2, n)} \leftrightarrow v_{(\alpha, n-1, n)}$ (where the second column-2 entry shifts from $n-2$ to $n-1$).

For $n \geq 9$, all axial distances in (c), (d) satisfy $|d| \geq 3$, hence the 2-blocks are non-degenerate (Hoefsmit coefficients in $\mathbb{Q}(q)^{\times}$).

Proof. Direct case-by-case verification using Lemma 1.4 of May-14 (positions in seminormal SYTs). For (a) with $(\alpha, \gamma) = (n-5, n-4)$, $\delta = n$: column-1 entries are $\{1\} \cup \{2, \dots, n\} \setminus \{n-5, n-4, n\}$, sorted $1, 2, \dots, n-6, n-3, n-2, n-1$; thus $n-2$ at row $n-4$, $n-1$ at row $n-3$, adjacent. Other cases similar.

For the swap-partner SYTs in (c), (d): e.g., in (c) with $(\alpha, \gamma) = (n-5, n-4)$, swapping $n-2$ and $n-1$ moves $n-1$ to row $n-4$ of column 1 and $n-2$ to $(3, 2)$, yielding SYT $(n-5, n-4, n-2)$, which is valid since $a_3 = n-2 > b_2 = 3$ for $n \geq 6$.

The axial distances follow from $c(i) = \text{col}(i) - \text{row}(i)$ applied to each case: e.g., $(c(n-1) - c(n-2))$ in (c) is $(-1) - ((1 - (n-4))) = (-1) - (5-n) = n-6$.

For $n \geq 9$: $|d| \geq |n-6| \geq 3$ and $|4-n| \geq 5$, both $\neq 1$. □

Proposition 12 (Image after $(T_{n-2}+1)$). $(T_{n-2}+1) \Phi(\hat{u}^{(\mu)})$ is a linear combination of the following 8 seminormal basis vectors, with all coefficients nonzero:

$$\begin{aligned} \text{from } w_\Phi(n-5, n-4) : & \quad v_{(n-5, n-4, n-1)}, \quad v_{(n-5, n-4, n-2)} \\ \text{from } w_\Phi(n-5, n-3) : & \quad v_{(n-5, n-3, n-1)}, \quad v_{(n-5, n-3, n-2)} \\ \text{from } w_\Phi(n-4, n-2) : & \quad v_{(n-4, n-2, n)}, \quad v_{(n-4, n-1, n)} \\ \text{from } w_\Phi(n-3, n-2) : & \quad v_{(n-3, n-2, n)}, \quad v_{(n-3, n-1, n)} \end{aligned}$$

The eight supports are pairwise disjoint.

Proof. For each of the 4 terms in (3), apply $(T_{n-2}+1)$ to the two-element seminormal expansion of $w_\Phi(\alpha, \gamma) = v_{(\alpha, \gamma, n-1)} + \tau v_{(\alpha, \gamma, n)}$. By Lemma 11:

Term 1: $w_\Phi(n-5, n-4)$. By Lemma 11(a), $(T_{n-2}+1)v_{(n-5, n-4, n)} = 0$. By Lemma 11(c), $(T_{n-2}+1)v_{(n-5, n-4, n-1)} = \kappa(\alpha' v_{(n-5, n-4, n-1)} + \beta' v_{(n-5, n-4, n-2)})$ with $\kappa, \alpha', \beta' \in \mathbb{Q}(q)^\times$ (Hoefsmit at $d = n-6$). The resulting support is $\{(n-5, n-4, n-1), (n-5, n-4, n-2)\}$, with the coefficient $A\kappa\alpha' \neq 0$ at the first SYT and $A\kappa\beta' \neq 0$ at the second.

Term 2: $w_\Phi(n-5, n-3)$. By Lemma 11(a), $(T_{n-2}+1)v_{(n-5, n-3, n)} = 0$. By Lemma 11(c), support of $(T_{n-2}+1)v_{(n-5, n-3, n-1)}$ is $\{(n-5, n-3, n-1), (n-5, n-3, n-2)\}$. Coefficients $B\kappa\alpha'$ and $B\kappa\beta'$, both nonzero.

Term 3: $w_\Phi(n-4, n-2)$. By Lemma 11(b), $(T_{n-2}+1)v_{(n-4, n-2, n-1)} = 0$. By Lemma 11(d), $(T_{n-2}+1)v_{(n-4, n-2, n)} = \kappa'(\alpha'' v_{(n-4, n-2, n)} + \beta'' v_{(n-4, n-1, n)})$ with $\kappa', \alpha'', \beta'' \in \mathbb{Q}(q)^\times$ (Hoefsmit at $d = 4-n$). Support $\{(n-4, n-2, n), (n-4, n-1, n)\}$. Coefficients $C\tau\kappa'\alpha''$ and $C\tau\kappa'\beta''$ (the τ from w_Φ), both nonzero.

Term 4: $w_\Phi(n-3, n-2)$. Same as Term 3 with $\alpha = n-3$. Support $\{(n-3, n-2, n), (n-3, n-1, n)\}$.

Disjointness. The 8 supports listed all have distinct (α, γ, δ) -triples (different middle column-2 entries or different positions of the boundary index), so they are pairwise disjoint. Hence the 8 coefficients are independent and all nonzero. $\square$

5.2 Step (b): apply $(T_{n-3}+1)$

Lemma 13 (Case analysis of T_{n-3}). For each of the 8 SYTs in Proposition 12:

- (a) $(n-5, n-4, n-1)$: $n-3$ at $(n-5, 1)$, $n-2$ at $(n-4, 1)$. Same column 1 adjacent. $T_{n-3}v = -v$.
- (b) $(n-5, n-4, n-2)$: $n-3$ at $(n-5, 1)$ (content $6-n$), $n-2$ at $(3, 2)$ (content -1). T_{n-3} acts as a 2-block at axial distance $d = n-7$, coupling $v_{(n-5, n-4, n-2)} \leftrightarrow v_{(n-5, n-4, n-3)}$.
- (c) $(n-5, n-3, n-1)$: $n-3$ at $(2, 2)$ (content 0), $n-2$ at $(n-4, 1)$ (content $5-n$). T_{n-3} acts as a 2-block at axial distance $d = 5-n$, coupling $v_{(n-5, n-3, n-1)} \leftrightarrow v_{(n-5, n-2, n-1)}$.
- (d) $(n-5, n-3, n-2)$: $n-3, n-2$ at $(2, 2), (3, 2)$ (same column 2 adjacent). $T_{n-3}v = -v$.
- (e) $(n-4, n-2, n)$: $n-3$ at $(n-4, 1)$ (content $5-n$), $n-2$ at $(2, 2)$ (content 0). T_{n-3} acts as a 2-block at axial distance $d = n-5$, coupling $v_{(n-4, n-2, n)} \leftrightarrow v_{(n-4, n-3, n)}$.
- (f) $(n-4, n-1, n)$: $n-3$ at $(n-4, 1)$, $n-2$ at $(n-3, 1)$. Same column 1 adjacent. $T_{n-3}v = -v$.
- (g) $(n-3, n-2, n)$: $n-3, n-2$ at $(1, 2), (2, 2)$ (same column 2 adjacent). $T_{n-3}v = -v$.
- (h) $(n-3, n-1, n)$: $n-3$ at $(1, 2)$ (content 1), $n-2$ at $(n-3, 1)$ (content $4-n$). T_{n-3} acts as a 2-block at axial distance $d = 3-n$, coupling $v_{(n-3, n-1, n)} \leftrightarrow v_{(n-2, n-1, n)}$.

For $n \geq 9$, all axial distances in (b), (c), (e), (h) satisfy $|d| \geq 2$, so 2-blocks are non-degenerate.

Proof. Direct case analysis using the position formulas for the seminormal basis of λ . Cases (a), (d), (f), (g) are “ $T_{n-3} = -1$ ” cases (same-column adjacency); cases (b), (c), (e), (h) are 2-blocks. For $n = 9$: $|n - 7| = 2$, $|5 - n| = 4$, $|n - 5| = 4$, $|3 - n| = 6$, all ≥ 2 .

For the swap-partner SYTs: e.g., in (b), swap $n-3$ and $n-2$ in $(n-5, n-4, n-2)$ moves $n-3$ to $(3, 2)$ and $n-2$ to $(n-5, 1)$, yielding $(n-5, n-4, n-3)$ — valid since $n-4 < n-3 < a_2 = n-4$ wait no: $a_2 = n-4$ and the new $a_3 = n-3$ satisfies $a_2 < a_3$ since $n-4 < n-3$. Validity of column 1: $b_2 = 3 < n-3$. Good.

In (c), swap moves $n-2$ to $(2, 2)$ and $n-3$ to $(n-4, 1)$, yielding $(n-5, n-2, n-1)$ — valid since $n-5 < n-2 < n-1$ and $b_1 = 2 < a_2 = n-2$.

In (e), swap moves $n-3$ to $(2, 2)$ and $n-2$ to $(n-4, 1)$, yielding $(n-4, n-3, n)$ — valid since $n-4 < n-3 < n$ and $b_1 = 2 < n-3$.

In (h), swap moves $n-2$ to $(1, 2)$ and $n-3$ to $(n-3, 1)$, yielding $(n-2, n-1, n)$ — valid since $n-2 < n-1 < n$ and $b_1 = 2 < n-2$. $\square$

Proposition 14 (Final support). $(T_{n-3}+1)(T_{n-2}+1)\Phi(\hat{u}^{(\mu)})$ has seminormal support exactly equal to the 8-element set $S_3^{(n)}$ of Theorem 3, with all 8 coefficients nonzero.

Proof. By Proposition 12, $(T_{n-2}+1)\Phi(\hat{u}^{(\mu)})$ has 8-element pairwise-disjoint support:

$$\begin{aligned} \text{supp}_1 &= \{(n-5, n-4, n-1), (n-5, n-4, n-2)\}, & \text{supp}_2 &= \{(n-5, n-3, n-1), (n-5, n-3, n-2)\}, \\ \text{supp}_3 &= \{(n-4, n-2, n), (n-4, n-1, n)\}, & \text{supp}_4 &= \{(n-3, n-2, n), (n-3, n-1, n)\}. \end{aligned}$$

Each supp_i has both coefficients nonzero.

Apply $(T_{n-3}+1)$:

supp_1 : $(n-5, n-4, n-1)$ killed (Lemma 13(a)); $(n-5, n-4, n-2)$ produces 2-block to $(n-5, n-4, n-3)$ (Lemma 13(b)). New support: $\{(n-5, n-4, n-2), (n-5, n-4, n-3)\}$, two nonzero coefficients.

supp_2 : $(n-5, n-3, n-1)$ produces 2-block to $(n-5, n-2, n-1)$ (Lemma 13(c)); $(n-5, n-3, n-2)$ killed (Lemma 13(d)). New support: $\{(n-5, n-3, n-1), (n-5, n-2, n-1)\}$.

supp_3 : $(n-4, n-2, n)$ produces 2-block to $(n-4, n-3, n)$ (Lemma 13(e)); $(n-4, n-1, n)$ killed (Lemma 13(f)). New support: $\{(n-4, n-2, n), (n-4, n-3, n)\}$.

supp_4 : $(n-3, n-2, n)$ killed (Lemma 13(g)); $(n-3, n-1, n)$ produces 2-block to $(n-2, n-1, n)$ (Lemma 13(h)). New support: $\{(n-3, n-1, n), (n-2, n-1, n)\}$.

Combining: the eight final supports are

$$\begin{aligned} &\{(n-5, n-4, n-3), (n-5, n-4, n-2), (n-5, n-3, n-1), (n-5, n-2, n-1), \\ &\quad (n-4, n-3, n), (n-4, n-2, n), (n-3, n-1, n), (n-2, n-1, n)\}, \end{aligned}$$

which is exactly $S_3^{(n)}$.

Nonzero coefficients. We track coefficients: each supp_i contributes two new SYTs with coefficients of the form (nonzero coeff of $(T_{n-2}+1)$ step) $\cdot \kappa'' \cdot (\alpha'''$ or $\beta''')$, where $\kappa'', \alpha''', \beta'''$ are the Hoefsmit coefficients at the T_{n-3} axial distance, all nonzero. The 8 resulting supports are pairwise disjoint (no two sub-supports overlap, by inspection of the lists). Hence the 8 coefficients are independent products of nonzero terms, all nonzero. $\square$

5.3 Proof of Theorem 3

Proof of Theorem 3. By Theorem 9, $B_{n-2}^{(\lambda)} V_\lambda = (T_{n-3}+1)(T_{n-2}+1)\Phi(\hat{u}^{(\mu)})$ up to scalar. By Proposition 14, this is nonzero and has support exactly $S_3^{(n)}$ with 8 nonzero coefficients. Hence $\dim B_{n-2}^{(\lambda)} V_\lambda = 1$.

The E_1^- containment is Corollary 10 (and is also immediate since every SYT in $S_3^{(n)}$ has minimum entry $\geq n-5 \geq 4 > 2$, so 2 is in column 1 and the basis vector lies in E_1^-). $\square$

6 Computational verification

The proof above is rigorous for all $n \geq 9$. We have additionally verified it computationally in symbolic $\mathbb{Q}(q)$ at $n = 9$ and at the specific rational $q = 7$ for $n \in \{9, 10, 11\}$ (the verification script is `~/projects/scratch/2026-05-15-a3-reduction/verify_reduction.py` and `verify_symbolic.py`).

At $n = 9$, the explicit form of the generator of $B_7^{(\lambda)} V_\lambda$ (in the seminormal basis, normalized with first nonzero coefficient set to 1) is, with all coefficients nonzero rational functions in q :

$$\begin{aligned} \text{coef } v_{(4,5,6)} &= q^9(q+1)^2/(q^2+1), \\ \text{coef } v_{(4,5,7)} &= q^8(q+1)^3/(q^2+1), \\ \text{coef } v_{(4,6,8)} &= q^7(q+1)(q^4+2q^3+3q^2+2q+1)/(q^2+1)^2, \\ \text{coef } v_{(4,7,8)} &= q^6(q+1)(q^3+2q^2+2q+1)/(q^2+1), \\ \text{coef } v_{(5,6,9)} &= q^6(q+1)(q^4+3q^3+4q^2+3q+1)/(q^4+q^3+q^2+q+1), \\ \text{coef } v_{(5,7,9)} &= q^5 \cdot (\text{nonzero in } \mathbb{Q}(q)), \\ \text{coef } v_{(6,8,9)} &= q^4 \cdot (\text{nonzero in } \mathbb{Q}(q)), \\ \text{coef } v_{(7,8,9)} &= q^3(q+1)^3. \end{aligned}$$

The support is exactly the 8 SYTs of $S_3^{(9)}$, as predicted.

The same 8-element support pattern (shifted by $n-5$) appears at $n = 10, 11$.

7 Discussion

7.1 The structural picture

Theorem 9 expresses $B_{n-2}^{(\lambda)} V_\lambda$ as a single $H_q(S_{n-2})$ -equivariant lift of $B_{n-3}^{(\mu)} V_\mu$ followed by $(T_{n-3}+1)(T_{n-2}+1)$. This is the representation-theoretic “ T_2 -obstruction” from the May-14 note turned around: the obstruction to a clean reduction (a at n) $\mapsto$ ($a-1$ at $n-2$) via simply forgetting the top row is now isolated to the two extra factors $(T_{n-3}+1)(T_{n-2}+1)$, which can be analyzed case-by-case.

7.2 Pattern for general a

Conjecturally, the same strategy extends to $\lambda = (2^a, 1^{n-2a})$ at $n \geq 3a$:

- Phase A: $R'_n V_\lambda = E_{n-1}^+ \cong V_{\mu_1}|_{n-2}$ via Φ , where $\mu_1 = (2^{a-1}, 1^{n-2a})$.
- Iterating Theorem 9-style commutations,

$$B_{n-a+1}^{(\lambda)} V_\lambda = \left[\prod_{k=0}^{a-2} (T_{n-a+1-k}+1) \right] \cdot \Phi^{(a)}(B_{\mu_{a-1}}^{(\mu_{a-1})} V_{\mu_{a-1}}),$$

where μ_{a-1} is the partition reached by iterated branching and $\Phi^{(a)}$ is the composite branching iso.

- The product of (T_j+1) factors performs the “L/R boundary choice” at each level, generating the 2^a support via the recursive structure S_a .

This is not yet proved. The key obstruction for general a is verifying that each successive branching iso $\Phi^{(k)}$ exists (needs the (T_j+1) -image after the k -th step to factor through a single $H_q(S_{n-2k})$ -isotypic component). At $a = 3$ the analogue holds because $\Phi(B_{n-3}^{(\mu)} V_\mu) \subseteq E_{n-3}^+ \cap E_{n-1}^+$ by Step 2 of Theorem 9, which is structurally a single isotypic copy.

7.3 The $\mathbb{Z}/2$ symmetry of S_a

The set S_3 (and generally S_a) is closed under the involution $x \mapsto (2a-1) - x$ (May-14 Section 3). This reversal, applied to the 8 SYTs of $S_3^{(n)}$, swaps:

$$\begin{aligned} (n-5, n-4, n-3) &\leftrightarrow (n-2, n-1, n), \\ (n-5, n-4, n-2) &\leftrightarrow (n-3, n-1, n), \\ (n-5, n-3, n-1) &\leftrightarrow (n-4, n-2, n), \\ (n-5, n-2, n-1) &\leftrightarrow (n-4, n-3, n). \end{aligned}$$

This is exactly the involution exchanging the “supp _{i} side” (post- $(T_{n-2}+1)$) of Proposition 12: $\text{supp}_1 \leftrightarrow \text{supp}_4$, $\text{supp}_2 \leftrightarrow \text{supp}_3$ after one $(T_{n-3}+1)$ application.

The representation-theoretic source: $\lambda = (2, 2, 2, 1^{n-6})$ has two “extremal” length-preserving corners (the column-2 corner $(3, 2)$ and the column-1 “bottom” which is conceptually adjacent), and the L/R choice corresponds to which corner is “filled first” as one fills cells from n downward.

8 Honest assessment

What is fully proved.

1. Theorem 3: the j -formula for $\lambda = (2, 2, 2, 1^{n-6})$ at every $n \geq 9$ is rigorously established, with support exactly $S_3^{(n)}$ (8 SYTs, all coefficients nonzero in $\mathbb{Q}(q)$).
2. Corollary 4: rank-zero $\Pi^{S_n}|_{V_{(2,2,2,1^{n-6})}} = 0$ for all $n \geq 9$, the second non-hook family fully structurally proved.
3. The branching reduction Theorem 9 is a clean structural statement, isolating the entire $a = 3$ proof to the May-13 $a = 2$ result plus two explicit Hoefsmit case analyses.

What is left open.

1. The general- a conjecture (May-14 Conjecture 5.1) remains conjectural. The reduction-via-branching strategy here suggests a path: iterate Theorem 9-style commutations, but verifying that each successive branching iso lands in a single isotypic component is nontrivial.
2. The combinatorial origin of S_a (Dyck-like paths? parking functions? Eulerian numbers?) is not identified. The $\mathbb{Z}/2$ reversal symmetry suggests palindromic combinatorics.

Status of the rank-zero program.

- Fully structurally proved: (1^n) , $(2, 1^{n-2})$, $(3, 1^{n-3})$ at $n \geq 5$, $(2, 2, 1^{n-4})$ at $n \geq 6$, and now $(2, 2, 2, 1^{n-6})$ at $n \geq 9$.
- Computationally verified: $(2^a, 1^{n-2a})$ at $a = 4, 5$ and $n \geq 3a$, support $= 2^a$.
- Conjectural: general $\lambda = (2^a, 1^{n-2a})$ at $n \geq 3a$.